ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 11



UNIT 4
Use of Multimedia

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

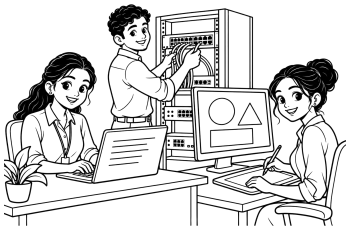
Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Explain pixels, bits per pixel, resolution and colour in digital graphics.
- [ ] Work out colours, bits per pixel, pixels and RGB / HTML colour codes.
- [ ] Compare lossy and lossless compression, and raster and vector graphics.
- [ ] Create, save and export graphics with layers in GIMP.
- [ ] Explain the types of frames and make 2D animations in Vectorian Giotto.
- [ ] Record, edit, save and export audio content with Audacity.
- [ ] Make, save and share a video clip with Windows Movie Maker.

HOW TO USE THIS BOOK

Read each chapter with your teacher, do the practical tasks and then write the exercises. The **WRITING SPACES** are for the notes your teacher gives you in class - write them neatly: they are your revision notes for the O/L examination.



THE WORDS YOU WILL LEARN

Digital graphic	Pixel	Bitmap	Resolution
Bits per pixel	RGB model	CMYK model	RGB Triplet
Lossy	Lossless	Raster graphic	Vector graphic
GIMP	Layer	Opacity	Animation
Key frame	Tween frame	Timeline	Work stage
Audacity	Audio track	Video clip	Transition

"A picture, a movement and a voice can say more than a page of words. Learn to use all three."

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"Do your best work today. Tomorrow you will be proud of it."

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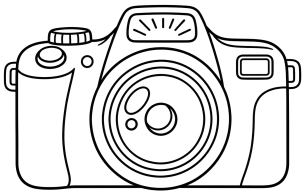
CHAPTER 1

Digital Graphics

A PHOTO FOR THE SCHOOL MAGAZINE

"My photo is sharp on my phone, but full of small squares on the big poster!"

In this chapter: pixels, colours, resolution, compression, raster and vector graphics.



4.1 Digital graphics and their basic elements

People have drawn and coloured pictures to share ideas since ancient times - old wall paintings show it. Today graphic software makes it simple to create and edit illustrations. Drawings or images created with computer graphic software are called **digital graphics**. Their four basic elements are the **pixel**, the **resolution**, the **size** and the **colour**.



Figure 4.1 - Basic elements of digital graphics

4.2 Pixels

A **pixel** is a tiny illuminated dot with a colour, displayed on a computer screen. Digital graphics are made of thousands of pixels in rows and columns. A digital graphic is an array of rectangular pixels called a **bitmap**. When a graphic is made bigger, the size of its pixels changes: a graphic with a small number of pixels loses quality when it is enlarged (Figures 4.2, 4.3).

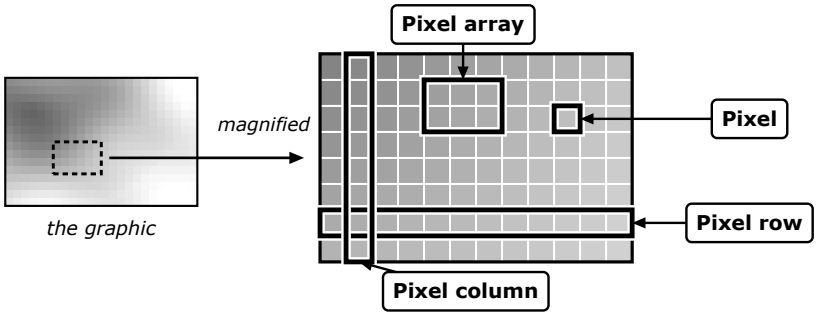


Figure 4.2 - Pixels

WRITING SPACE 1 - Digital graphics write what your teacher tells you

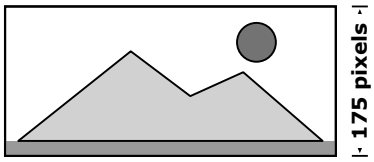
4.3 Colours and bits per pixel

The colours of a graphic are stored with a number of **bits per pixel (bpp)**: one bit gives two colours (0, 1), two bits give four (00, 01, 10, 11). More bits per pixel give a higher quality. **Number of colours = 2^{bpp}**: 4 bits give 2 × 2 × 2 × 2 = 16 colours. Backwards, 16 colours = 2^4, so they need 4 bits per pixel.

Bits per pixel	Number of colours	Bits per pixel	Number of colours
1 bpp (2^1)	2	7 bpp (2^7)	128
2 bpp (2^2)	4	8 bpp (2^8)	256
3 bpp (2^3)	8	10 bpp (2^{10})	1,024
4 bpp (2^4)	16	16 bpp (2^{16})	65,536
5 bpp (2^5)	32	24 bpp (2^{24})	16,777,216
6 bpp (2^6)	64	32 bpp (2^{32})	4,294,967,296

WRITING SPACE 2 - Colours and bits per pixel — write what your teacher tells you

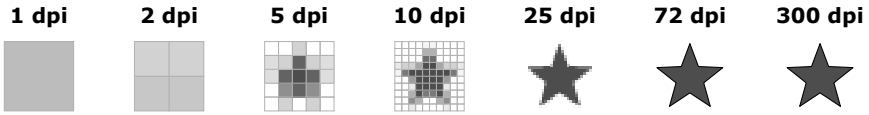
4.4 Resolution



(43,750 pixels)

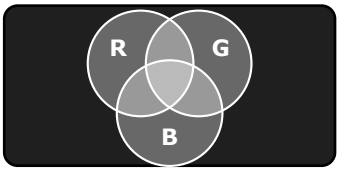
Figure 4.4 - Image resolution

Pixels measure the physical dimension of a graphic, shown as the **image resolution**. Figure 4.4 is 250 pixels wide and 175 pixels high: 250 × 175 = **43,750 pixels**. A high resolution graphic is higher in quality. Quality is also given in **pixels per inch (ppi)** or **dots per inch (dpi)** (Figure 4.5).

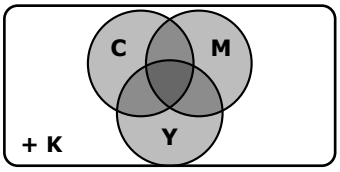


4.5 Colour

Each pixel represents a colour; about **16 million** colours are visible to the naked eye. The **colour model** suits the output device: **RGB** (red, green, blue light) for screens, **CMYK** (cyan, magenta, yellow, black ink) for printing. **Primary colours** are single colours; **secondary** mix two, **tertiary** mix three. A primary colour has **256 variations (0 - 255)**.



RGB model - lights
screens



CMYK model - inks
printing on paper

"Every great picture on a screen is made of tiny dots. Small things build big things."

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A tertiary colour is a combination from 000, 000, 000 to 255, 255, 255: the **RGB Triplet**, e.g. RGB (245, 102, 36). In hexadecimal it is RGB (F5, 66, 24): each value 0 - 255 is two hexadecimal digits. GIMP shows it as the **HTML notation**: R 143, G 28, B 25 is **8f1c19** (Figure 4.9).

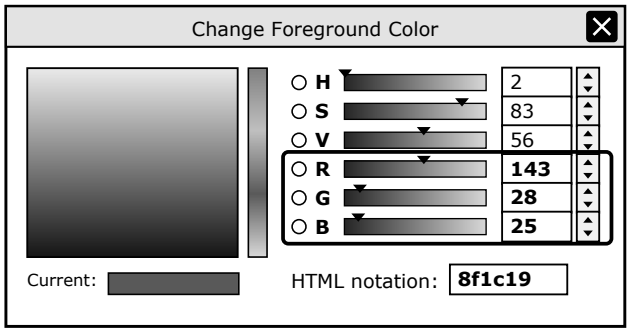


Figure 4.9 - GIMP: Change Foreground Color

4.6 Graphic size and compression

The size of a graphic depends on its pixels, lines and colour combinations. Many pixels, many colours and a high resolution give a large file that is hard to store and send. **Compression** makes it smaller, when it is saved or later. **Lossy** formats make the smallest files but reduce the quality - faster saving and downloading (JPEG). **Lossless** formats keep the original quality (GIF, PNG, RAW).

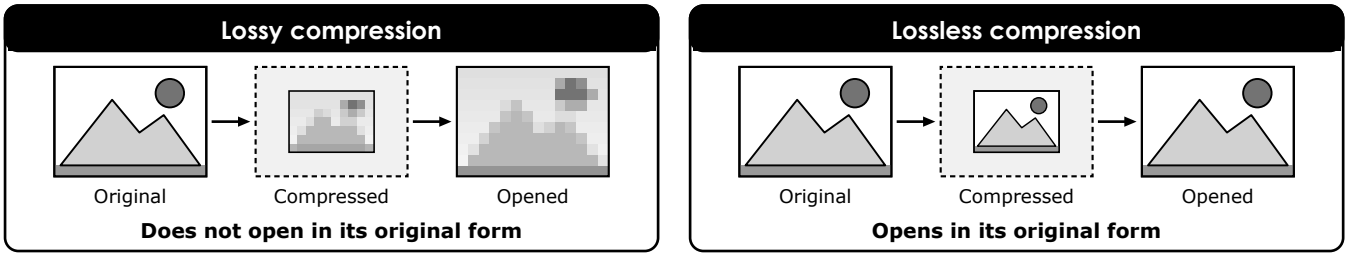


Figure 4.10 - Lossy and lossless compression

Size before compression: bits = pixels × bits per pixel (8 bits = 1 byte). 640 × 400 pixels with 24 bits per pixel: 256,000 × 24 ÷ 8 = **768,000 bytes**.

4.7 Raster and vector graphics

Digital graphics are **raster** or **vector graphics**; the software decides the type (Figure 4.11).

Criteria	Raster graphic	Vector graphic
Creation of the graphic	An array of pixels in different colours	A collection of straight or curved lines
File records	No records	A record of the start and the end, the number of lines, straight and curved lines and the colours used
Quality of the graphic	Quality is lost when the size changes	Quality is not lost when the size changes
High quality creations	Not suitable	Suitable
Creation and saving	Uses less memory space	Needs more memory space
Computer speed in creation	Does not change	Reduces
Software (examples)	Adobe Photoshop, GIMP, Corel PHOTO-PAINT, Artweaver, Microsoft Photo Editor	Adobe Illustrator, CorelDRAW, DrawPlus, Microsoft Expression Design, Xara Photo & Graphic Designer

"Look closer and you will see the pixels. Look further and you will see the picture."

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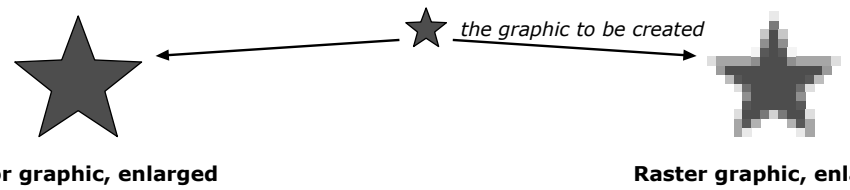


Figure 4.11 - A star enlarged as a vector and as a raster graphic

WRITING SPACE 3 - Lossy and lossless — write what your teacher tells you

DO IT - PRACTICAL 1 (TEXTBOOK ACTIVITY, PAGE 123, ITEMS 5 AND 6)

Pixels, colour codes and compression. **Ask your teacher first.**

1	Open a photograph in GIMP (File → Open). Zoom in with the **Magnify** tool (Z) until you can see the pixels.
2	Click the foreground colour: type R 143, G 28, B 25 and read the **HTML notation**. Try 000000 and ffffff.
3	Export it as a **JPEG** and a **PNG** file (File → Export). Compare the sizes and the quality: which is lossy? Is a photograph raster or vector?

A Bits and colours

8 marks

Fill the empty cells. Colours = 2 to the power of the bits per pixel.

Bits per pixel	Number of colours	Bits per pixel	Number of colours
1			256
3		10	
	16		65,536
6		24	

B Work it out

6 marks

Pixels = width × height. Bytes = pixels × bits per pixel ÷ 8. Work on the side.

Graphic	Width × height	Number of pixels	Bits per pixel	Size in bytes
Figure 4.4 (p. 84)	250 × 175		-	-
A new image in GIMP (p. 95)	1200 × 800		24	
The graphic with text (p. 100)	640 × 400		8	
The same graphic, 24 bits	640 × 400	256,000	24	

C RGB and HTML notation

8 marks

Change each value (0 - 255) to two hexadecimal digits, or back. The first row is done.

Colour (textbook page)	R	G	B	HTML notation
Example (p. 86)	245	102	36	f56624
Figure 4.9	143	28	25	
Gradient colour 1 (p. 100)				29c89c
Gradient colour 2 (p. 100)				e3216a
Black (p. 95)	0	0	0	

D Which one?

8 marks

Lossy or lossless compression? Raster or vector graphic? Tick one box.

It is ...	Lossy	Lossless	Raster	Vector
The file is as small as possible, but the quality is reduced.	☐	☐	☐	☐
It is made of an array of pixels in different colours.	☐	☐	☐	☐
It keeps a record of its lines, their start and end, and the colours.	☐	☐	☐	☐
The graphic opens again in its original form.	☐	☐	☐	☐
JPEG	☐	☐	☐	☐
Its quality is lost when its size changes.	☐	☐	☐	☐
GIF and PNG	☐	☐	☐	☐
It is made of a collection of straight or curved lines.	☐	☐	☐	☐

"Quality is never an accident. It is the result of care in every step."

- THE KING OF ICT

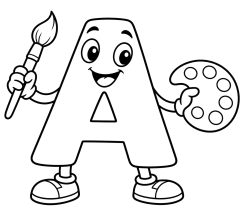
CHAPTER 2

Graphic Design with GIMP

? A BANNER FOR SPORTS DAY

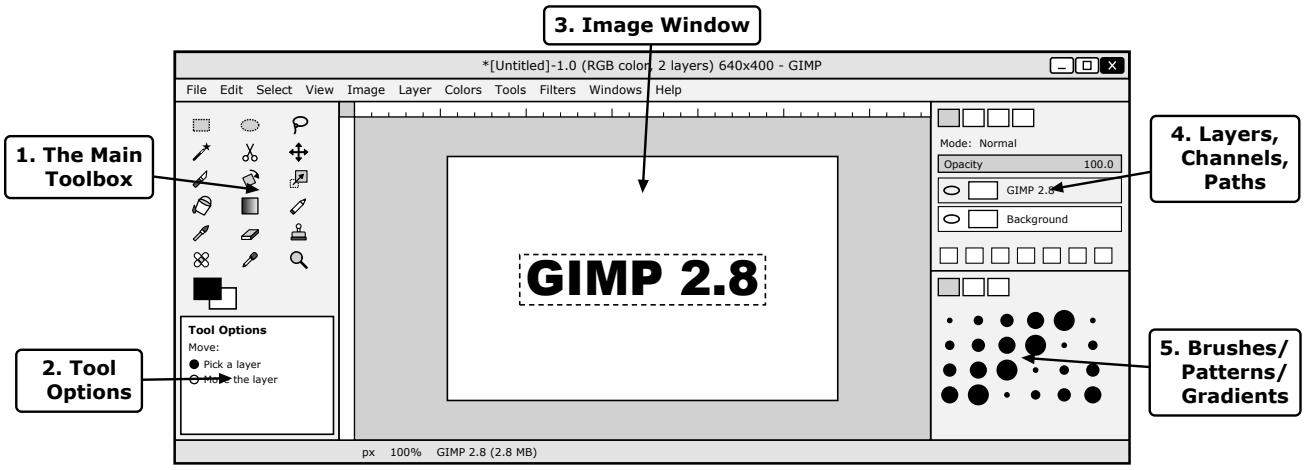
"We need a banner for Sports Day, but we cannot buy software."
"Use GIMP: it is free!"

In this chapter: the GIMP window and tools, layers, and graphics from photographs and text.



4.8 GIMP and its window

Most graphic software must be bought, and some does not work on every operating system. **GIMP** is a free image manipulation program created for the GNU free and open source operating system, to reconstruct photographs and to create, edit and format graphics. It produces **raster graphics**. It is UNIX based and also runs on Microsoft Windows and Mac (www.gimp.org/downloads/).



The GIMP 2.8 window (textbook page 89)

1 The Main Toolbox: the tools for editing graphics. Show or hide them: Edit → Preferences → Toolbox; **2 Tool Options**: customise the options of the selected tool; **3 Image Window**: shows the graphic. Windows → Single-Window Mode keeps everything in one window; **4 Layers, Channels, Paths**: tabs: click one to open it; **5 Brushes/Patterns/Gradients**: tabs that help create and add colour to a graphic.

4.9 The Main Toolbox

Selection, brush, transform and other tools (pages 90 - 92); the letters are the shortcut keys:

Rectangle (R) Selects a rectangle.	**Ellipse** (E) Selects a circle or an ellipse.	**Free (Lasso)** (F) Free-form selections.	**Fuzzy** (U) Magic Wand: similar colours.
Bucket Fill (Shift+B) A colour or a pattern.	**Blend** (L) Fills with a gradient.	**Pencil** (N) Paints hard-edged lines.	**Paintbrush** (P) Paints soft-edged lines.
Eraser (Shift+E) Erases pixels from a layer.	**Heal** (H) Removes unwanted spots.	**Move** (M) Moves a selection.	**Crop** (Shift+C) Crops or clips the image.
Scale (Shift+T) Scales a layer or a selection.	**Colour Picker** (O) Picks a colour of an image.	**Magnify** (Z) Alters the zoom level.	**Text** (T) Places text into the image.

Also in the toolbox: By Colour (Shift + O), Scissors (I), Foreground, Airbrush (A), Ink (K), Clone (C), Perspective Clone, Convolve (Blur/Sharpen) (Shift + U), Smudge (S), Dodge/Burn (Shift + D), Align (Q), Rotate (Shift + R), Shear (Shift + S), Perspective (Shift + P), Path (B), Measure (Shift + M).

> "Software is only a tool. The ideas and the patience come from you."
>
> - THE KING OF ICT

4.10 New, save, export and open

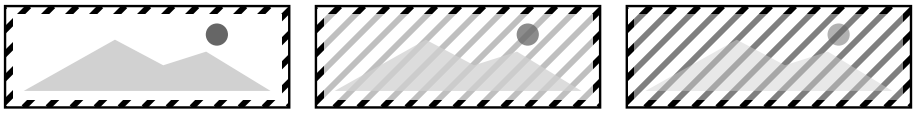
Task	What to do (pages 93 - 94)
A new graphic	File → New. In **New Image** set the width and the height, or choose a template. Click OK.
Save	File → Save. Type a name in **Save Image**, choose the place from **Places**, click Save. GIMP saves in **XCF** (e.g. image1.xcf).
Export	File → Export. Type a name, choose the place and the file type in **Select File Type (By Extension)**, click Export. The graphic is compressed to suit the file type.
Open	File → Open. Choose the place from **Places**, select the graphic in **Names**, click Open.

WRITING SPACE 4 - Save and export

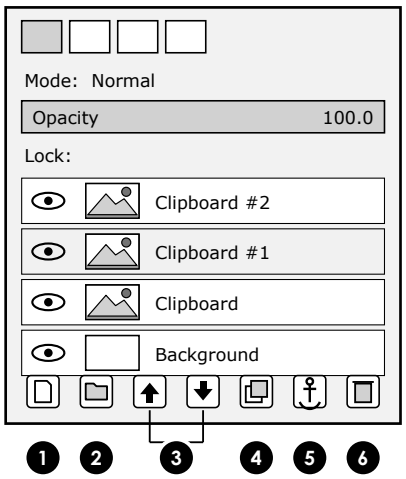
write what your teacher tells you

4.11 Layers

Layers are like sheets of transparent paper: an object on an upper layer may hide the layer below. Use a separate layer for each picture, so a change on one layer (text, colour, shapes, editing) does not affect another. The **eye** makes a layer visible or invisible. The buttons: **1** New, **2** Group, **3** Move (up / down), **4** Duplicate layer, **5** Anchor, **6** Delete. **Opacity** changes how much of the layers below can be seen:



Opacity 100% **Opacity 75%** **Opacity 50%**



WRITING SPACE 5 - Layers

write what your teacher tells you

WRITING SPACE 6 - GIMP

write what your teacher tells you

4.12 Making graphics with GIMP

Pictures can be scanned, or taken with a digital camera or a smart phone camera and saved in the computer. The two graphics of the textbook:

A FROM THREE PICTURES (PP. 94 - 99)

1. Windows → Single-Window Mode; open the pictures one at a time (not Open as Layers).
2. File → New: 1200 × 800 px (black and white: 000000, ffffff).
3. Right click the layer of a picture: **Add an Alpha Channel**.
4. Ellipse tool; Select → Feather 10 pixels; Edit → Copy Visible.
5. Edit → Paste as → New Layer; Move tool; Scale 450 × 600.
6. Repeat for the other pictures; **Merge Visible Layers**.
7. Crop; Filters → Decor → Add Border (10 px blue, 15 px yellow).
8. Save, then export.

B WITH TEXT (PP. 100 - 102)

1. File → New: 640 × 400 px.
2. Colours **29c89c** and **e3216a**; drag the Gradient tool from left to right.
3. Text tool: Gill Sans Ultra Bold, 72, black: **NATURE**.
4. A layer **Text Border** below the text; **Alpha to Selection**.
5. Select → Grow 2 pixels; white Bucket Fill; Select → None.
6. Filters → Light and Shadow → Drop Shadow.
7. Merge Visible Layers, border, save, export.

DO IT - PRACTICAL 2 (TEXTBOOK ACTIVITY, PAGE 102)

Graphic design with GIMP. **Ask your teacher first.**

1	Make a graphic from photographs you took on a special occasion (4.12 A). Save it (.xcf) and export it.
2	Give it a better finish with other **Filters**.
3	With your teacher, download pictures of special places in Sri Lanka and make graphics with titles.
4	Make an invitation for digital printing (4.12 B) and a banner for a school event.

A Complete the menu paths

8 marks

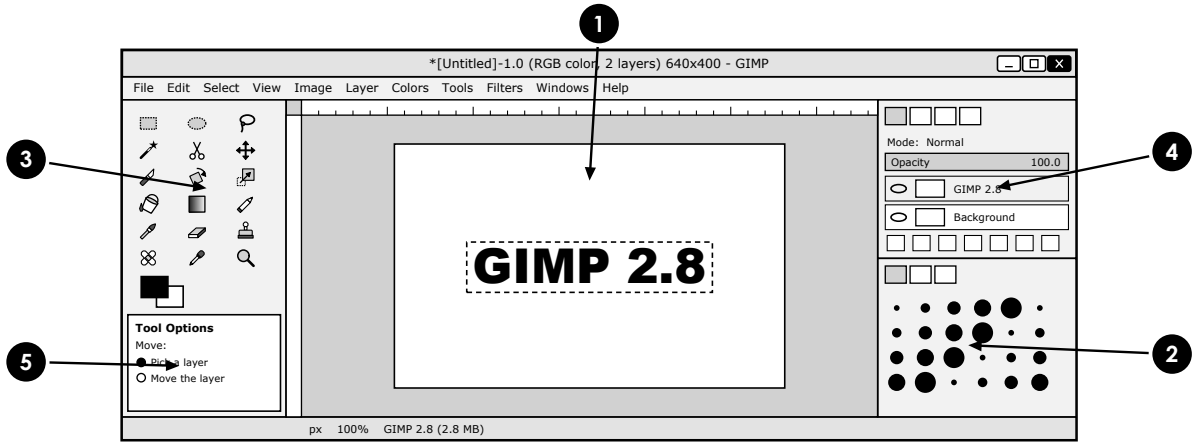
Write the menu path you use in GIMP.

You want to ...	Menu path
keep GIMP in one window	
show or hide the tools of the toolbox	
make the edge of a selection soft	
paste a copied part as a new layer	
make a selection 2 pixels bigger	
add a border to a graphic	
add a shadow to text	
save the graphic as a JPEG or PNG file	

B Label the GIMP window

5 marks

Write the name of each numbered part.



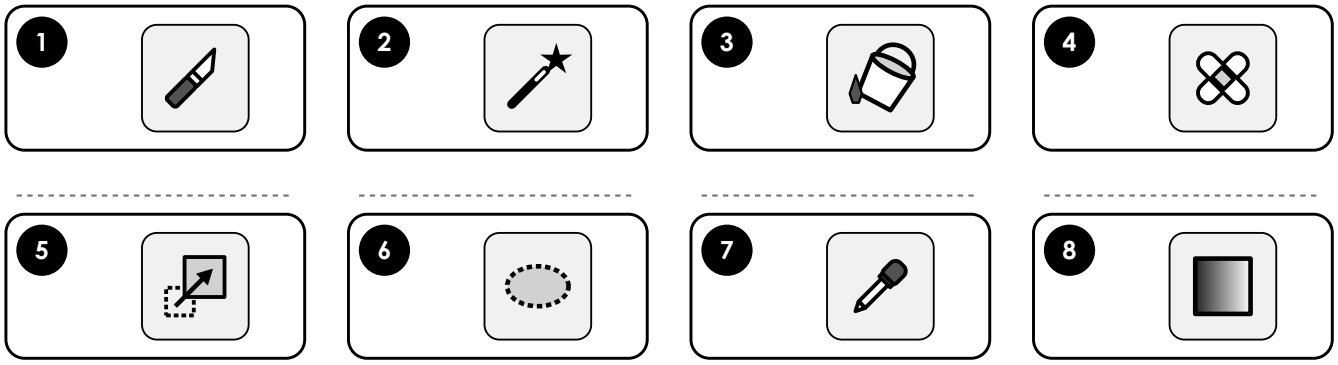
1. ..... 2. ..... 3. .....

4. ..... 5. .....

C Which tool?

8 marks

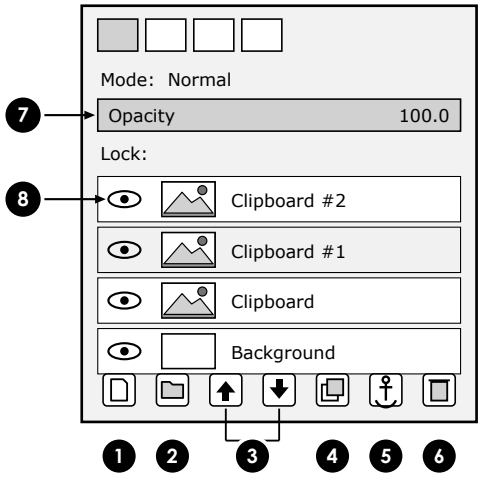
Write the name of each tool and its shortcut key.



D The Layers window

8 marks

Write what each numbered part is.



1. .....
2. .....
3. .....
4. .....
5. .....
6. .....
7. .....
8. .....

CHAPTER 3

Two-dimensional Animation

A CARTOON FOR SCIENCE DAY

"Can my ball bounce?" "Draw two key frames - the software makes the rest!"

In this chapter: animation, the types of frames, and 2D animations with Vectorian Giotto.



4.13 What is animation?

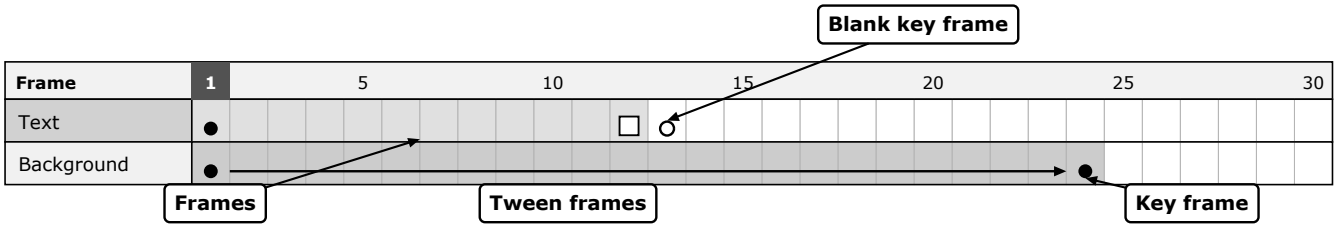
The main aim of animation is to give more depth to a static graphic and make it look real. **Animation** is an optical illusion that shows the movements of objects: a series of frames is arranged in sequential order and displayed continuously. Changing the speed of the frames changes the speed of the objects. Animations are used in commercial publications, to promote products, in computer games and in cartoon films.

WRITING SPACE 7 - Animation write what your teacher tells you

4.14 Types of frames

Frame	What it is (page 103)
Key frame	The main, special static frame of an animation. The user decides it. E.g. the first and the last frames of a motion.
Tween frame	Made by the software between two frames, for a smooth motion.
Frame	Added next to a frame with content: the content stays longer in the animation.
Blank key frame	An empty frame to put new content in. With content, it is no longer a blank key frame.

Smooth motion uses **24 frames per second (fps)**. With key frames at the beginning and at the end of a motion of 24 frames, the software creates the other **22 tween frames**.

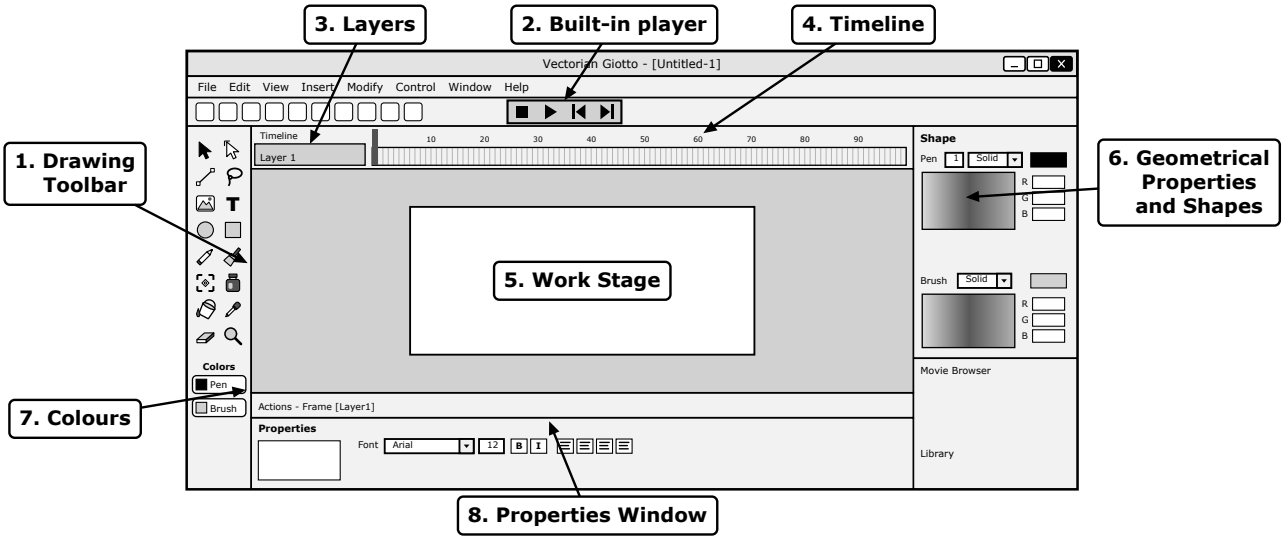


The frames on the Timeline of Vectorian Giotto (pages 111 - 113)

WRITING SPACE 8 - Types of frames write what your teacher tells you

4.15 Vectorian Giotto and its window

Vectorian Giotto is free animation software (vectorian.com/giotto). It is easy to use and needs no coding or complex scripts; it embeds motion into created images, has nearly 50 flash animation tools and makes simple animations with music.



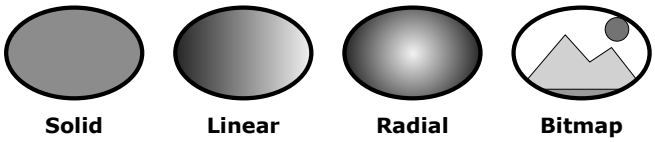
The Vectorian Giotto window (textbook page 104)

1 Drawing Toolbar: tools to create objects and text on the work stage; **2 Built-in player**: plays, stops, rewinds or moves to the end of the animation; **3 Layers**: like transparent papers. Right click: add or remove a layer, show or hide it, lock it; **4 Timeline**: sets the time period of the animation; **5 Work Stage**: animations are created here; **6 Geometrical Properties and Shapes**: colours of shapes, border type and colour, colour combinations, designs and images; **7 Colours**: pen = the border colour; Brush = the fill colour; **8 Properties Window**: font type, font size and alignment of the text.

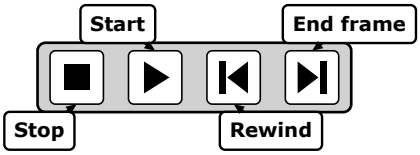
4.16 Drawing tools, shapes and colours

Line (N) Draws straight lines.	**Insert bitmap** (M) Adds a picture.	**Text** (T) Adds text.	**Oval** (O) Draws ovals.
Rectangle (R) Draws rectangles.	**Pencil** (P) Draws free lines.	**Brush** (B) Draws wider shapes.	**Selection** (V) Selects or moves objects.
Sub selection (A) Changes a shape by its marks.	**Lasso** (L) Cuts and removes shapes.	**Free transform** (Q) Resizes, rotates, reshapes.	**Ink bottle** (S) Changes the border.
Paint Bucket (K) Changes the fill colour.	**Eyedropper** (I) Copies colours to an object.	**Eraser** (E) Erases a section only.	

Pen gives the border: the line type (Solid, Dash, Dot, DashDot, DashDotDot), the colour and the density. **Brush** gives the fill: one colour (Solid), a combination of colours (Linear, Radial) or a picture (Bitmap).



The **Built-in player** plays, stops, rewinds or moves to the end of the animation. Insert bitmap (M) adds bmp, jpg, tif, png, gif, ico pictures.



4.17 New, save, export and open

Task	What to do (page 110)
A new animation	File → New Movie
Save	File → Save: a name, a place, Save. Giotto saves in the **.vgd** format.
Export for the web	File → Export Flash Movie: a place and a name, Save; in **Export Settings** select **Compress movie**, OK. The file is a **.swf** file.
Open	File → Open. To play a .swf file, right click it and select your web browser.

4.18 Making animations

Modify → Document opens **Document properties**: the frame rate in fps (the speed), the background colour and the dimensions, e.g. 550 × 400 px at 12 fps.

A A SIMPLE ANIMATION (PP. 111 - 112)

1. Open Giotto. Draw a square and an oval at the top of the work stage.
2. Give them border and fill colours with the **Pen** and the **Brush**. This is the first key frame.
3. Right click frame 24 on the Timeline: **Insert Key frame**. Drag the shapes to the bottom.
4. Right click a frame between 1 and 24: **Create Motion Tween**. Play it with the Built-in player.

B WITH CAPTIONS (PP. 112 - 113)

1. Draw a rectangle; Brush → Linear; fill it from the colour basket.
2. Name Layer 1 **Background**; right click, **+ Layer**, name it **Text**.
3. On the Text layer type **Beautiful Sri Lanka** (Curlz MT, 35) and centre it.
4. Right click the text: Effect → Add; select **Falling Leaf**, OK.
5. Background layer: right click frame 83 (the end of the text) and insert a key frame. Save, export, open.

DO IT - PRACTICAL 3 (TEXTBOOK ACTIVITY, PAGE 113)

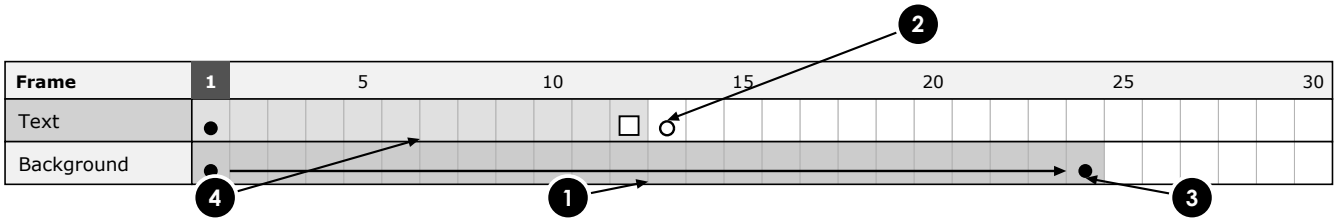
A motion text on your own image. **Ask your teacher first.**

1	File → New Movie; put your GIMP graphic on the work stage (Insert bitmap, M).
2	Name Layer 1 **Background**; add a layer (+ Layer) named **Text**.
3	Type a caption on the Text layer (T); right click it, Effect → Add, choose an effect.
4	Insert a key frame on the Background layer at the end of the text. Play it.
5	Save (.vgd), export (File → Export Flash Movie) and open the .swf file in a browser.

A Name the frames

4 marks

Write the type of each numbered frame.



1. ..... 2. ..... 3. ..... 4. .....

B Frames and seconds

4 marks

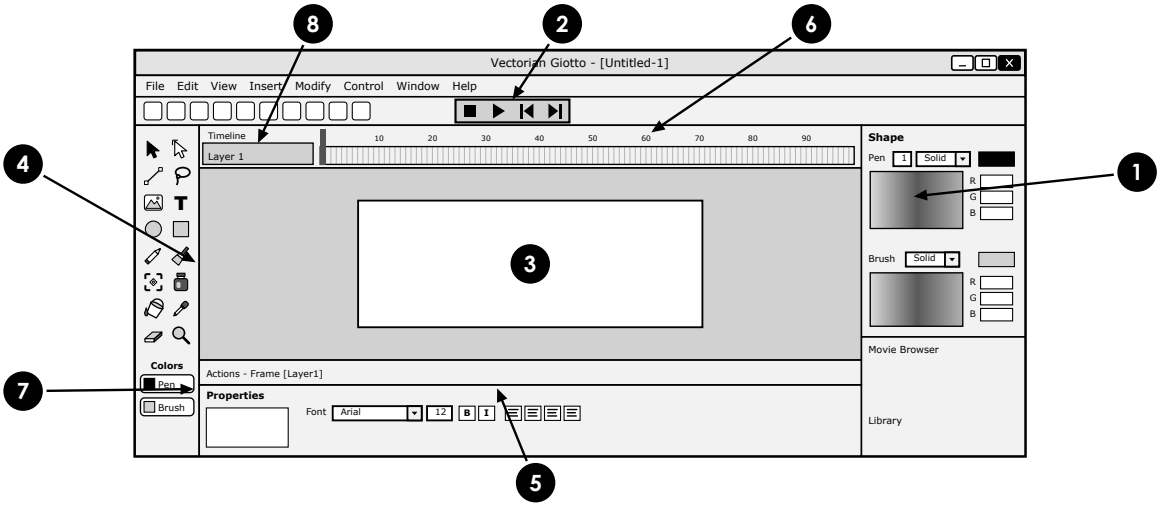
Work out each answer.

1. Frames 1 and 24 are key frames. How many tween frames? .....
2. At 24 fps, how many frames are there in 3 seconds? .....
3. At 12 fps, how many seconds do 24 frames take? .....
4. 12 fps is changed to 24 fps. What happens to the speed? .....

C Label the Giotto window

8 marks

Write the name of each numbered part.



1. ..... 2. ..... 3. ..... 4. .....

5. ..... 6. ..... 7. ..... 8. .....

WRITING SPACE 9 - Uses of animation

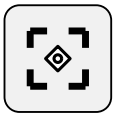
write what your teacher tells you

D Which tool?

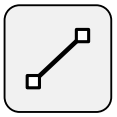
8 marks

Write the name of each Giotto tool and its shortcut key.

1 

2 

3 

4 

5 

6 

7 

8 

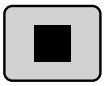
E Buttons and fills

8 marks

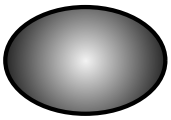
Boxes 1 - 4: name the Built-in player button. Boxes 5 - 8: name the Brush fill.

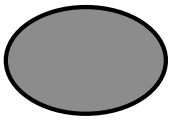
1 

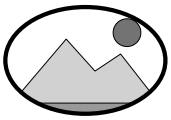
2 

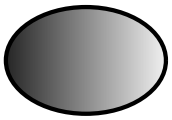
3 

4 

5 

6 

7 

8 

WRITING SPACE 10 - Frames per second

write what your teacher tells you

CHAPTER 4

Audio and Video

A VIDEO ABOUT SIGIRIYA

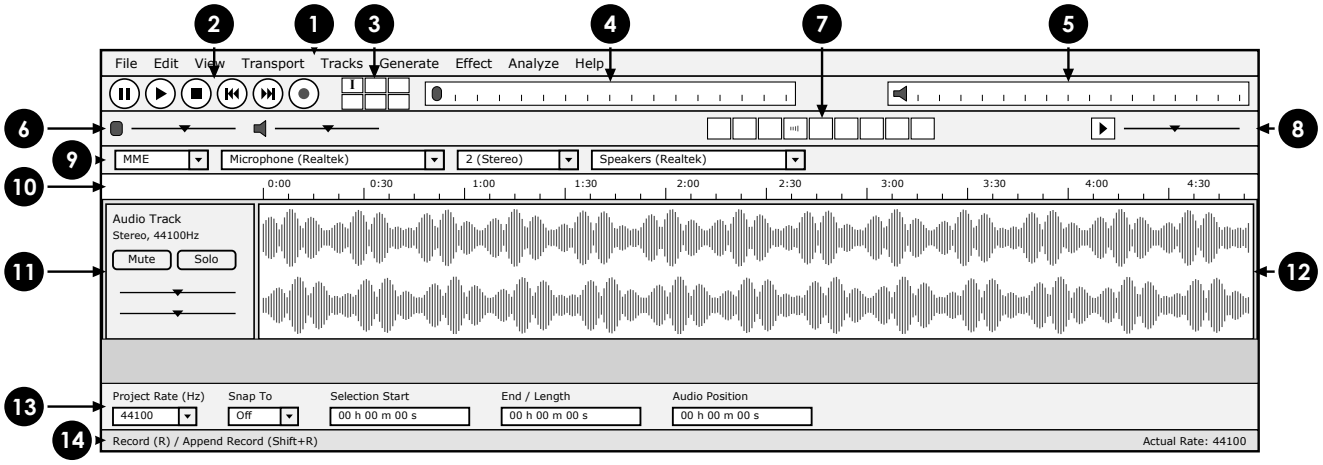
"Let's join our Sigiriya photos and our voices into one video!"

In this chapter: recording and editing audio, and making and sharing a video clip.



4.19 Audio content and Audacity

Voice or sound makes graphics and animations more meaningful and attractive. Files created with recorded sounds or voices are **audio content**. Audio software: Audacity, Power Sound Editor, Mp3DirectCut, Music Editor Free, Wavosaur, Ardour, WavePad Sound Editor, Sound Engine. **Audacity** is free software for Windows, Mac and Linux, for recording and editing, with multi layers (audacity.sourceforge.net). Its functions: recording live audio proceedings; recording music being played in the computer; inserting sound effects into recordings for CDs / DVDs; editing WAV, AIFF, FLAC, MP2, MP3 or Ogg Vorbis files; copying, trimming, mixing or joining sounds; changing the speed and the pitch of recordings.

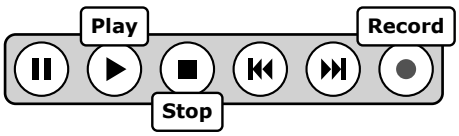


The Audacity window (textbook page 115)

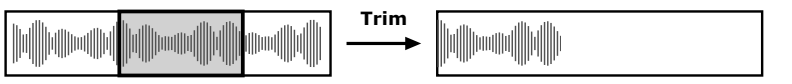
1 Menu Bar
2 Transport Toolbar
3 Tools Toolbar
4 Recording Meter Toolbar
5 Playback Meter Toolbar
6 Mixer Toolbar
7 Edit Toolbar
8 Transcription Toolbar
9 Device Toolbar
10 Timeline
11 Track Control Panel
12 Audio Track
13 Selection Toolbar
14 Status Bar

4.20 Record, save, export and edit

Task	What to do (pages 116 - 117)
A new window	File → New
Record	Click **Record** on the Transport Toolbar and use the microphone; **Stop**; **Play** to listen.
Save	File → Save. Audacity saves an **.aup** file (Audacity Project) that can be edited.
Export	File → Export Audio: the format in **Save as type** (WAV, AIFF, FLAC, MP2, MP3); the audio is compressed. **Edit Metadata**: the name, the year, the title ...
Trim	Open the .aup file. Select the part you need with the **Selection Tool**; click **Trim Audio** (Edit Toolbar); save.



The Transport Toolbar



1 Select the part you need **2 Trim Audio: the rest is removed**

Trim Audio (page 117)

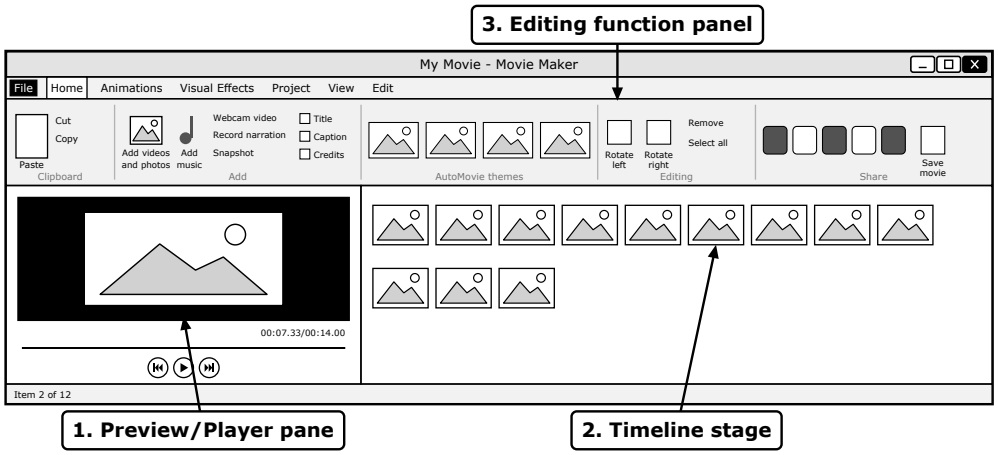
> "Tell your story with pictures, sound and movement - and tell it well."
>
> - THE KING OF ICT

WRITING SPACE 11 - Audacity

write what your teacher tells you

4.21 Video clips and Windows Movie Maker

A video created with a series of visuals is a **video clip**: from a recorded video, or still pictures and audio content. Software: PhotoBucket, YouTube Remixer, Movie Masher, One True Media, Motion Box, Stash Space, Windows Movie Maker, AVI Edit, Super DVD Video Editor. **Windows Movie Maker** is free software from Microsoft to create and edit video clips (www.windows-movie-maker.org).



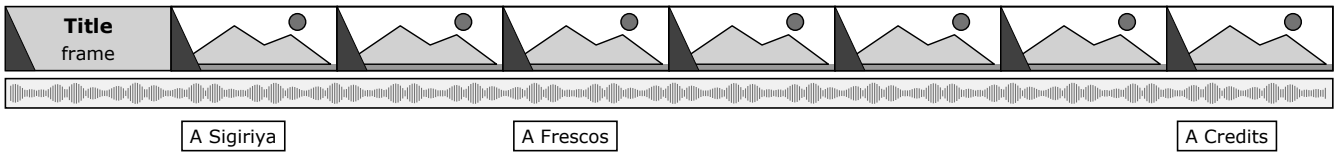
The Windows Movie Maker window (textbook page 118)

	Part	What it does
1	**Preview/Player pane**	To view the videos and image frames, and to watch the clip before saving.
2	**Timeline stage**	The video clips, the image frames and the audio clips are arranged here.
3	**Editing function panel**	Tools to edit video and audio content and to add visual effects.

4.22 Making a video clip

Add the video clips and photos. A **transition** shows the relationship between two image frames and how the frames appear: open **Animations**, click the first frame, point at each transition, then click one. Play the clip in the Preview/Player pane and edit it.

You want to ...	Use
add video clips and photos	Home → Add videos and photos
add a transition between image frames	the **Animations** menu
set the time an image frame is shown	Animations → Duration
use the same effect for all image frames	**Apply to all**
make an image frame move	**Pan and Zoom**
add a topic at the start	Home → Title
add text to an image frame	Home → Caption
introduce the direction, artists, music and venue	Home → Credits
add a song or music	Home → Add music



The Timeline stage at the end: transitions, a title, captions, credits and music (page 121)

4.23 Saving and sharing a video clip

File → Save Project saves a **.wlmp** file (Movie Maker Project) that can be edited later. To watch the clip on a computer, a television, a mobile phone or on social media (Facebook, YouTube, Flickr), use File → Save Movie: choose the medium, a place and a name, then Save.

File, Save Movie: the setting	For ...
For email	to send it by e-mail
For computer	to watch it on a computer
For high-definition display	a high-definition display
Android Phone (large / medium)	mobile phones
YouTube, Facebook, Vimeo, Flickr, OneDrive ...	social media
Audio only	as an audio content

WRITING SPACE 12 - Saving a video — write what your teacher tells you

DO IT - PRACTICAL 4 (TEXTBOOK ACTIVITY, PAGE 123, ITEMS 1 - 4)

A video clip about places in Sri Lanka. **Ask your teacher first.**

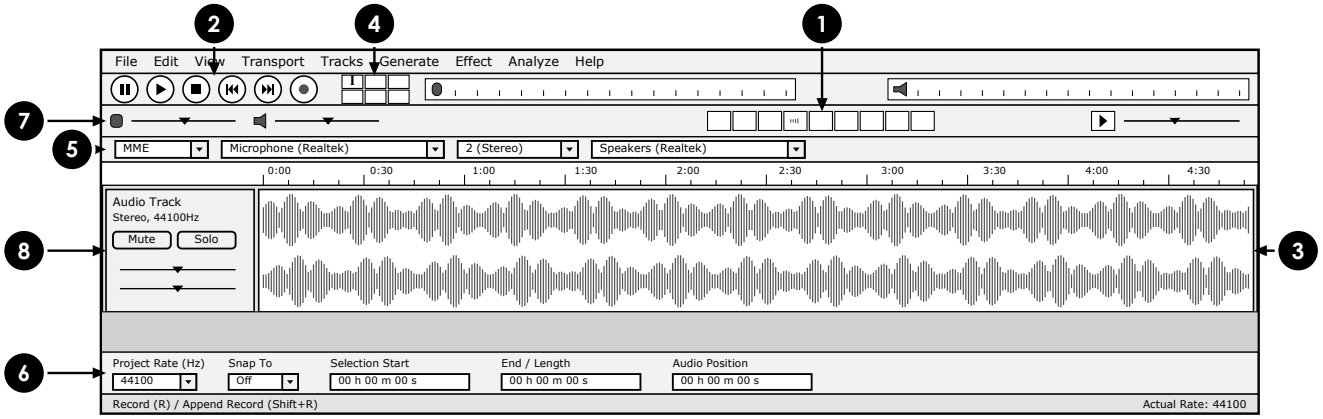
1	Make still graphics of special places of interest in Sri Lanka (GIMP) and a few 2D animations with a still graphic in the background (Giotto).
2	Record an audio content to suit them in Audacity. Trim it, save it (.aup) and export it as MP3.
3	In Windows Movie Maker make a video clip about the places with transitions, a title, captions, credits and music.
4	Save the project (.wlmp), then File → Save Movie (For computer). Show it to your class.

WRITING SPACE 13 - Audio software — write what your teacher tells you

A Label the Audacity window

8 marks

Write the name of each numbered part.



1. ..... 2. ..... 3. ..... 4. .....

5. ..... 6. ..... 7. ..... 8. .....

B Audio or video software?

8 marks

Tick one box for each program.

Software	Audio	Video
Wavosaur	☐	☐
Movie Masher	☐	☐
Audacity	☐	☐
AVI Edit	☐	☐
Ardour	☐	☐
Windows Movie Maker	☐	☐
WavePad Sound Editor	☐	☐
Super DVD Video Editor	☐	☐

C Which command or setting?

8 marks

Write the Windows Movie Maker command (1 - 6) or the File, Save Movie setting (7 - 8).

You want to ...	Command or setting
1. add video clips and photos	
2. add a transition between image frames	
3. set the time an image frame is shown	
4. make an image frame move	
5. add a topic at the start	
6. introduce the direction, artists, music and venue	
7. send the clip to a friend by e-mail	
8. put the clip on YouTube or Facebook	

REVISION

Unit 4 in One Page

Read this the night before the test

Topic	The key facts
Digital graphics	Drawings or images made with graphic software. Basic elements: pixel, resolution, size, colour.
Pixel, bitmap	A pixel is a tiny dot with a colour. A bitmap is an array of rectangular pixels.
Bits per pixel	Colours = 2^{bpp}: 1 bit 2, 8 bits 256, 24 bits 16,777,216 colours.
Resolution	Width × height in pixels: 250 × 175 = 43,750. Quality: ppi or dpi.
Colour	About 16 million colours. RGB: lights, screens. CMYK: inks, printing. Primary, secondary (2), tertiary (3).
RGB Triplet	0, 0, 0 to 255, 255, 255. RGB (245, 102, 36) = F5 66 24 in hexadecimal.
Compression	Lossy: smallest file, loses quality (JPEG). Lossless: keeps quality (GIF, PNG, RAW).
Raster / vector	Raster: pixels, loses quality when resized. Vector: lines, keeps quality.
GIMP	Free, open source, raster. Toolbox, Tool Options, Image Window, Layers, Brushes. XCF; File, Export.
Layers	Transparent sheets. Eye: show / hide. Opacity. New, Group, Move, Duplicate, Anchor, Delete.
Animation	An optical illusion: frames shown in order. Key frames (user), tween frames (software), 24 fps.
Vectorian Giotto	Free, no coding. Toolbar, player, layers, Timeline, Work Stage ... Save .vgd; Export Flash Movie .swf.
Audacity	Records and edits audio. Record, Stop, Play; Trim Audio. Save .aup; Export Audio: WAV ... MP3.
Movie Maker	Add videos and photos, transitions, title, caption, credits, music. Save Project .wlmp; Save Movie.
Project files	GIMP .xcf (export JPEG, PNG); Giotto .vgd (export .swf); Audacity .aup (export WAV, MP3 ...); Movie Maker .wlmp.

TICK WHEN YOU CAN DO IT

- ☐ I can explain pixels, bits per pixel, resolution and colour models.
- ☐ I can work out colours, bits, pixels and RGB / HTML colour codes.
- ☐ I can compare lossy and lossless compression, and raster and vector graphics.
- ☐ I can make, save and export a graphic with layers in GIMP.
- ☐ I can explain the types of frames and make a 2D animation in Giotto.
- ☐ I can record, trim, save and export audio content with Audacity.
- ☐ I can make, save and share a video clip with Windows Movie Maker.

WORD POWER

Unit 4 Word List

Twenty-six words you must know

Word	What it means
Digital graphic	A drawing or image created with computer graphic software.
Pixel	A tiny illuminated dot with a colour on a computer screen.
Bitmap	A digital graphic: an array of rectangular pixels.
Bits per pixel (bpp)	The bits used for one pixel. Colours = 2^{bpp}.
Resolution	The size of a graphic in pixels: width × height.
ppi / dpi	Pixels per inch / dots per inch: they show the quality of a graphic.
RGB model	Red, green and blue light: television and computer screens.
CMYK model	Cyan, magenta, yellow and black inks: printing on paper.
Primary colour	A single colour. It has 256 variations (0 - 255).
Secondary colour	A colour made by mixing two colours.
Tertiary colour	A colour made by mixing three colours.
RGB Triplet	Three values from 0, 0, 0 to 255, 255, 255, e.g. RGB (245, 102, 36).
Lossy	Compression that makes the file smallest but reduces the quality.
Lossless	Compression that keeps the original quality.
Raster graphic	A graphic made of pixels; it loses quality when resized.
Vector graphic	A graphic made of lines; it keeps its quality when resized.
GIMP	Free, open source image manipulation program (raster graphics).
Layer	Like a sheet of transparent paper for one part of a graphic.
Opacity	How much a layer hides the layers below it.
Animation	An optical illusion: a series of frames shown in order.
Key frame	A main static frame that the user creates.
Tween frame	A frame the software makes between two frames.
Timeline	It sets the time period of an animation.
Audio content	Files created with recorded sounds or voices.
Video clip	A video created with a series of visuals.
Transition	It shows how one image frame changes to the next.

"Learn the words today, and you will speak like an expert tomorrow."

- THE KING OF ICT

WRITING SPACE 14 - Multimedia in this unit

write what your teacher tells you

END OF UNIT

Unit 4 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. The four basic elements of a digital graphic are pixel, resolution, size and
   (a) sound (b) colour (c) layer (d) frame
2. How many colours can a pixel of 8 bits show?
   (a) 8 (b) 16 (c) 256 (d) 1024
3. The quality of a graphic is given in pixels per inch (ppi) or in
   (a) dpi (b) fps (c) MP3 (d) XCF
4. Graphics created in GIMP are saved in the
   (a) XCF format (b) JPEG format (c) GIF format (d) PNG format
5. Animation is
   (a) an optical illusion (b) a file format
   (c) a colour model (d) a sound
6. A Vectorian Giotto animation for web designing is exported in the
   (a) .vgd format (b) .swf format (c) .xcf format (d) .wlmp format
7. In Windows Movie Maker, clips and image frames are arranged on the
   (a) Timeline stage (b) Player pane (c) Ribbon (d) Status Bar
8. A - Trim Audio keeps the selected part. B - A Movie Maker project is saved as .wlmp. C - Lossy compression keeps the quality. Which is correct?
   (a) A only (b) B only (c) A and B (d) B and C

2 Fill in the blanks

8 marks

One word or a few words in each blank.

1. A digital graphic is an array of rectangular pixels called a ..... .
2. The ..... model uses cyan, magenta, yellow and black inks.
3. Colours made by mixing three colours are ..... colours.
4. Add an Alpha ..... to a layer to make a picture transparent.
5. In Giotto the border colour comes from the Pen and the fill colour from the ..... .
6. Animations are created on the ..... of Vectorian Giotto.
7. Audacity projects are saved in the ..... file format.
8. In Windows Movie Maker a ..... shows the relationship between two image frames.

3 Name the pictures

8 marks

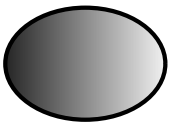
Cards 1 - 2: raster or vector? 3 - 4: GIMP tool. 5 - 6: Brush fill. 7 - 8: the button.

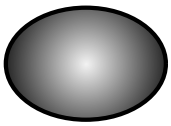
1 

2 

3 

4 

5 

6 

7 

8 

4 Answer these

16 marks

Write in the space given. Show your working.

1. A graphic uses 16 bits per pixel. How many colours can it show? How many bits per pixel do 32 colours need? (2 marks)

2. A graphic is 640 pixels wide and 400 pixels high. Find the number of pixels, and its size in bytes with 24 bits per pixel. (2 marks)

3. Write two differences between lossy and lossless compression. (2 marks)

4. Write two differences between raster graphics and vector graphics. (2 marks)

5. Write two uses of layers in GIMP. (2 marks)

6. What is the difference between a key frame and a tween frame? (2 marks)

7. Write two functions of Audacity. (2 marks)

8. Name two settings of File, Save Movie in Windows Movie Maker and say what each is for. (2 marks)